

1. Administrative & Study Identification

Full Study Title: An Open-label, Randomized, Parallel Group, Non-inferiority Study to Investigate the Pharmacokinetics, Pharmacodynamics, Safety and Tolerability of a New Maintenance Dosing Regimen of Ofatumumab, Followed by Extended Treatment in Participants With Relapsing Multiple Sclerosis **Short Title/ Acronym:** FILIOS **Protocol Number:** COMB157Q12301 **NCT Number:** NCT06869785 **Posting ID:** 938400358219260155 **Sponsor Name:** Novartis (Company: Novartis Pharmaceuticals) **Investigational Product:** Ofatumumab (anti-CD20) **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Status Description:** Currently recruiting participants **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (area under the curve [AUC] up to 12 weeks), pharmacodynamics, safety, and tolerability of a new maintenance dosing regimen/dosage of ofatumumab compared to the approved dosage. **Secondary Objectives:** To evaluate long-term safety, tolerability, and extended treatment outcomes in participants with relapsing multiple sclerosis.

2.2 Background & Rationale To address the need for an optimized maintenance dosing regimen of ofatumumab in patients with Relapsing Multiple Sclerosis (RMS). This study will generate data to compare the pharmacokinetic and pharmacodynamic profiles, as well as the long-term safety, of a new dosage of ofatumumab against the currently approved standard dose to improve clinical practice and potentially patient convenience.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Approved Dosage) **Blinding:** Open-label **Randomization:** Randomized, Parallel Group

3.2 Planned Enrollment & Duration Target \$N\$: 180 participants **Number of Sites:** 64 locations **Estimated Study Start:** 13-Mar-2025 **Estimated Primary Completion:** 11-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed informed consent must be obtained prior to participation in the study. 2. Male or female study participants aged 18 to 60 years (inclusive) at screening. 3. Diagnosis of multiple sclerosis (MS) according to the 2017 Revised McDonald criteria (Thompson et al 2018). Relapsing forms of MS: relapsing-remitting MS (RRMS), or active secondary progressive MS (SPMS).

4.2 Exclusion Criteria 1. Participants suspected of not being able or willing to cooperate or comply with study protocol requirements in the opinion of the Investigator or emergence of any clinically significant condition/disease (e.g. active systemic bacterial, viral or fungal infections) during screening prior to Day 1 which might result in safety risk for participants. 2. Participants with a history of confirmed progressive multifocal leukoencephalopathy (PML) or neurological symptoms consistent with PML. 3. Participants at risk of developing or having reactivation of hepatitis. 4. Emergence of active chronic disease (or stable but treated with immune therapy) prior to Day 1 of the immune system other than MS (e.g. rheumatoid arthritis, scleroderma, Sjögren's syndrome, Crohn's disease, ulcerative colitis, etc.) or with immunodeficiency syndrome (hereditary immune deficiency, drug-induced immune deficiency). 5. Pregnant or nursing (lactating) women. 6. History of lymphoproliferative disease or any known malignancy or history of malignancy of any organ system (except for basal cell carcinoma, or squamous cell carcinomas of the skin that have been treated with no evidence of recurrence in the past 3 months). 7. Participants taking prohibited therapies, including B cell targeted therapies (e.g. such as ocrelizumab, rituximab, ofatumumab, ublituximab, and inebilizumab).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product: Ofatumumab (Drug Preferred Name: ofatumumab) **Trade Names:** Arzerra, Kesimpta **ATC Codes:** L01FA02, L04AA52 **EMA URLs:** - https://www.ema.europa.eu/en/documents/product-information/arzerra-epar-product-information_en.pdf - https://www.ema.europa.eu/en/documents/product-information/kesimpta-epar-product-information_en.pdf **Dosage/Frequency:** Arm 1: Active Comparator Ofatumumab dose 1 (Approved dosage); Arm 2: Experimental Ofatumumab dose 2 (New dosage). **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** Initial 12 weeks for primary endpoint PK assessment, followed by an extended treatment duration until study completion (estimated March 2030).

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant therapies allowed by the protocol that do not interfere with immune response. **Prohibited:** B cell targeted

therapies (e.g. ocrelizumab, rituximab, ofatumumab prior to trial baseline, ublituximab, and inebilizumab).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -X to 0	Informed Consent, Medical History, McDonald Criteria Evaluation
Baseline	Day 0	Randomization, First Dose
Interim	Weeks 1 to 12	Pharmacokinetic (PK) tracking (AUC), PD Assessment, Safety Labs
End of Study	Mar 2030	Final Vital Signs, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Ofatumumab plasma pharmacokinetics - area under the curve, up to 12 weeks.
Measurement Method: Central Laboratory Analysis of plasma samples.

7.2 Secondary & Exploratory Endpoints 1. Ofatumumab plasma pharmacokinetics - minimum observed plasma concentration. 2. Ofatumumab plasma pharmacokinetics - maximum observed plasma concentration. 3. Incidence of Adverse events (AE) and Serious adverse events (SAE). 4. Additional evaluation of pharmacodynamics, overall safety, and tolerability profiles over the primary treatment period and the extended long-term follow-up phase.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection evaluating clinical and safety laboratory parameters. Emphasis on monitoring for infections (such as PML) and injection-related reactions.
Serious Adverse Events (SAEs): Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External committee details as managed by Novartis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for effectiveness outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N = 180$ powered to demonstrate non-inferiority of the new dosage regimen versus the approved dose regarding PK/PD targets. **Handling of Missing Data:** Standard multiple imputation or Last Observation Carried Forward (LOCF) methodologies per Novartis statistical protocols.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring per Novartis clinical operations plan. **Audit Trail:** eCRF locking and data clarification forms.

11. Study Identifiers & Governance

Primary ID: {{COMB157Q12301}} **Registry Number:** {{NCT06869785}}
Sponsor/Lead Organization: {{Novartis Pharmaceuticals}} **Study Title:** {{FILIOS}} **Official Regulatory Title:** {{An Open-label, Randomized, Parallel Group, Non-inferiority Study to Investigate the Pharmacokinetics, Pharmacodynamics, Safety and Tolerability of a New Maintenance Dosing Regimen of Ofatumumab, Followed by Extended Treatment in Participants With Relapsing Multiple Sclerosis}} **Source Level:** {{5}}

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: {{Relapsing Multiple Sclerosis (RMS)}}
Condition Name: {{Multiple Sclerosis}} **Preferred Name:** {{Multiple Sclerosis}} **Preferred UMLS Name:** {{Multiple sclerosis, relapsing remitting}} **Semantic Type:** {{Disease or Syndrome}} **MeSH Code:** {{D009103}} **SNOMED ID:** {{24700007}} **Definition:** {{A progressive autoimmune disorder affecting the central nervous system resulting in demyelination. Patients develop physical and cognitive impairments that correspond with the affected nerve fibers.}} **Diagnosis Threshold:** {{2017 Revised McDonald criteria}}
Medical Methodology: {{B-cell targeted therapy (anti-CD20)}}
Optimization Target: {{Pharmacokinetic (AUC) non-inferiority}}

13. Study Procedures & Methodology

Management Pathway: {{Disease-modifying therapy for RMS}}
Education Protocol (HCP): {{Protocol administration and PK sampling training}} **Education Protocol (Patient):** {{Treatment compliance and safety monitoring}} **Quality Audit Mechanism:** {{Central laboratory PK/PD tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 260 weeks (Extended Treatment)}} **Onsite Visit Cadence:** {{Baseline, intensive PK sampling to Week 12, followed by extension schedule}} **Remote Monitoring Frequency:** {{Continuous safety monitoring}} **Checklist Review Interval:** {{Protocol defined intervals}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					